
EE 66
Fall 2026

Signals, Dynamics, and Information

Discussion 5A

1. How Many Frequencies Are There?

For an integer k , define the vector $\psi_k \in \mathbb{C}^p$ by

$$\psi_k[n] = e^{i\frac{2\pi}{p}kn}, \quad n = 0, 1, \dots, p-1.$$

Note that k is allowed to be *any* integer here, not just $0, 1, \dots, p-1$.

(a) Show that $\psi_{k+p} = \psi_k$. How many distinct vectors are there in the set $\{\psi_k : k \in \mathbb{Z}\}$?

(b) Show that $\psi_{p-k} = \psi_k^*$, where ψ_k^* denotes the entrywise complex conjugate of ψ_k .

- (c) Let $p = 6$. What are ψ_0 , ψ_3 , and ψ_5 ? Interpreted as a 6-periodic DT signal, which has the highest frequency?

2. DTFS and DFT, Its Fraternal Twin [Adapted from EE 120 Homework]

To represent finite-length or periodic discrete-time signals, engineers have traditionally used a complex exponential Fourier series expansion that is slightly different from the Discrete-Time Fourier Series (DTFS). The expansion of choice is called the Discrete Fourier Transform (DFT).

In this problem, you will discover the simple relationship between the DTFS and the DFT. Recall that the analysis and synthesis equations of the DTFS expansion of a periodic discrete-time signal $x : \mathbb{Z} \rightarrow \mathbb{C}$ are:

$$x[n] = \sum_{k=0}^{p-1} X_k e^{ik\omega_0 n} \quad \xleftrightarrow{\text{DTFS}} \quad X_k = \frac{1}{p} \sum_{n=0}^{p-1} x[n] e^{-ik\omega_0 n}$$

where $\omega_0 = 2\pi/p$ is the fundamental frequency of the signal.

- (a) Consider the set of basis signals $\{\boldsymbol{\varphi}_0, \dots, \boldsymbol{\varphi}_{p-1}\}$, where $\boldsymbol{\varphi}_k = \frac{1}{p} \boldsymbol{\psi}_k$. Determine $\langle \boldsymbol{\varphi}_k, \boldsymbol{\varphi}_l \rangle$ for each of the cases $k = l$ and $k \neq l$.

Hint: What do you know about $\langle \boldsymbol{\psi}_k, \boldsymbol{\psi}_l \rangle$?

(b) We wish to express a p -periodic signal x in terms of the basis functions in $\{\boldsymbol{\varphi}_0, \dots, \boldsymbol{\varphi}_{p-1}\}$, as follows:

$$x = \sum_{k=\langle p \rangle} X'_k \boldsymbol{\varphi}_k.$$

Determine an expression for the coefficients X'_k in terms of projections of the signal x onto the basis functions $\boldsymbol{\varphi}_k$.

- i. What is the relationship between the DFT coefficients X'_k and the DTFS coefficients X_k ?
- ii. Write down the analysis and synthesis equations for the DFT.